**38** When $\alpha$-particles are directed at gold leaf:

1 almost all $\alpha$-particles pass through without deflection

2 a few $\alpha$-particles are deviated through large angles.

What are the reasons for these effects?

| | 1 | 2 |
|---|---|---|
| **A** | most $\alpha$-particles have enough energy to pass right through the gold leaf | gold is very dense so a few low energy $\alpha$-particles bounce back from the gold surface |
| **B** | most $\alpha$-particles miss all gold atoms | a few $\alpha$-particles bounce off gold atoms |
| **C** | the gold nucleus is very small so most $\alpha$-particles miss all nuclei | occasionally the path of an $\alpha$-particle is close to a nucleus |
| **D** | the positive charge in an atom is not concentrated enough to deflect an $\alpha$-particle | occasionally an $\alpha$-particle experiences many small deflections in the same direction |